

S O C I N C I D E N T R E P O R T

Phishing Email Analysis

AgentTesla Malspam Campaign

Organisation Globex Manufacturing Ltd.	Incident Date 4 December 2024
Classification Phishing-based Malware Infection	Overall Risk Rating HIGH
Report Phase Phase 1: Static Email & Attachment Analysis	Report Author: Ukah Paul

T H R E A T I D E N T I F I E D

AgentTesla Infostealer — Credential Harvesting & Keylogging Malware

Delivered via spear-phishing attachment using file masquerading (MITRE T1036.008). Payload compiled and delivered same day. All email authentication controls absent or failed.

SECTION 1 | SUMMARY OF FINDINGS

On 4 December 2024, a staff member at Globex Manufacturing Ltd. received a targeted malspam email impersonating a Turkish supplier (Acron Su ve Çevre Teknolojileri A.Ş.). The email carried a malicious attachment disguised as a technical specification document. Static analysis of the email and attachment confirms delivery of the AgentTesla infostealer.

Finding	Detail
Attack Type	Targeted malspam — spear phishing via malicious attachment
Sender	sertan@acronas.com.tr — originating IP 94.141.120.32
Authentication Failures	SPF softfail, DKIM none, DMARC none — all three controls absent or failed
Social Engineering	Purchase quotation lure targeting manufacturing industry; real supplier branding used
Attachment	TECHNICAL SPECIFICATIONS.TAR — declared as TAR, actually RAR archive containing .exe
Payload	TECHNICAL SPECIFICATIONS.exe — PE32 .NET assembly (1.05 MB), consistent with AgentTesla
Key Evasion Technique	File masquerading (T1036.008) — .TAR extension conceals RAR archive hiding Windows executable
Confirmed Impact	Credential theft and keylogging confirmed by Phase 2 network (PCAP) analysis
Overall Severity	HIGH

SECTION 2 | EMAIL ANALYSIS RESULTS

2.1 Tools Used

Tool	Purpose
Sublime Text	Opened raw .eml file as plain text; inspected full header block before the first blank line
MX Toolbox Header Analyzer	Pasted headers to evaluate SPF, DKIM, DMARC, relay chain, and spoofing indicators
emldump.py	Listed all MIME parts; extracted email body and attachment from the .eml file
whois (Kali Linux / RIPE DB)	Queried RIPE NCC for ownership, registration, and abuse contact of originating IP 94.141.120.32

2.2 Header Analysis Results

Header Field	Value / Finding	Significance
From (display name)	Sertan ÇOKER (decoded from RFC 2047 Base64)	Turkish name — consistent with claimed identity
From (address)	sertan@acronas.com.tr	Sending domain is acronas.com.tr
Return-Path	sertan@acronas.com.tr	Matches From — no separate bounce redirection
Reply-To	(not set)	No reply diversion — entirely attachment-based attack
Originating IP	94.141.120.32	Direct single-hop connection — no relay anonymization
Sender Timestamp	4 Dec 2024 04:51:17 -0800 (US Pacific)	Timezone mismatch — Turkish sender using US Pacific offset; suggests automated infrastructure
Received Timestamp	Wed, 04 Dec 2024 12:51:16 UTC	Recipient mail server receipt time
Subject	PURCHASE QUOTATION	Generic business subject — low suspicion for manufacturing company
SPF	softfail	Sending IP not explicitly authorized — possible compromised account or bulk mailer
DKIM	none	No digital signature — message integrity cannot be verified
DMARC	none	No enforcement policy — spoofed mail from this domain passes uninspected

Email Overview in Sublime Text — Full Raw .eml Header Block

```
2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
1 Return-Path: <sertan@acronas.com.tr>
2 Delivered-To: [recipient's email address]
3 Authentication-Results: [recipient's email server]; dkim=none; dmarc=none; spf=softfail ([recipient's email server]: 94.141.120.32 is
  neither permitted nor denied by domain of sertan@acronas.com.tr) smtp.mailfrom=sertan@acronas.com.tr
4 Received: from [94.141.120.32] (unknown [94.141.120.32])
5   by [recipient's email server] (Postfix) with ESMTP id [information removed]
6   for <[recipient's email address]>; Wed, 04 Dec 2024 12:51:16 +0000 (UTC)
7 From: =?UTF-8?B?U2VydGFuIMOHT0tFUG==?<sertan@acronas.com.tr>
8 To: [recipient's email address]
9 Subject: PURCHASE QUOTATION
10 Date: 4 Dec 2024 04:51:17 -0800
11 Message-ID: <20241204045117.4A43A7B93A5F2488@acronas.com.tr>
12 MIME-Version: 1.0
13 Content-Type: multipart/mixed;
14   boundary="-----_NextPart_000_0012_34AB1E64.CFC22211"
15
16 This is a multi-part message in MIME format.
17
18 -----_NextPart_000_0012_34AB1E64.CFC22211
19 Content-Type: text/html
20 Content-Transfer-Encoding: quoted-printable
```

Email Header Analysis — MX Toolbox (SPF / DKIM / DMARC / Relay Chain Results)

SPF and DKIM Information

dmARC:acronas.com.tr

Hide

Solve Email Delivery Problems

Test	Result	
DMARC Record Published	No DMARC Record found	More Info

Reported by ns43.ns.tr on 5/27/2026 at 10:54:33 AM (UTC 0), just for you.

spf:acronas.com.tr:94.141.120.32

Hide

Solve Email Delivery Problems

Test	Result	
SPF Record Published	No SPF Record found	More Info

Reported by ns44.ns.tr on 5/27/2026 at 10:54:34 AM (UTC 0), just for you.

Session Transcript:

- - txt:acronas.com.tr

1 ns44.ns.tr 185.7.0.5 AUTH 140 ms Received 1 Referrals , rcode=NAME_ERROR com.tr. 3600 IN SOA mname=ns73.ns.tr rname=hostmaster.trabis.gov.tr serial=1779876830,

- WebServer [spf:acronas.com.tr:94.141.120.32] 281ms

RESTful API

Relay Information

Received

0 seconds

Delay:

From unknown to email

0

0.2

0.4

0.6

0.8

1

1.2

Hop

Delay

From

By

With

Time (UTC)

Blacklist

1

*

unknown 94.141.120.32

email

ESMTP

12/4/2024 12:51:16 PM

2.3 Originating IP Intelligence — WHOIS Analysis (94.141.120.32)

THREAT INTELLIGENCE FINDING

The sending IP resolves to a UK-based bulletproof-style hosting provider (DGTL TECH UK LLP) with a subnet created just 7 days after the attack. This is the third geographic mismatch in the evidence chain — the attacker claimed to be Turkish, used a US Pacific Timezone, and sent from UK-hosted infrastructure.

WHOIS data was retrieved from the RIPE NCC database for originating IP 94.141.120.32. The following intelligence was extracted and analyzed:

WHOIS Field	Value	SOC Analysis / Significance
IP / Subnet	94.141.120.0/24 (94.141.120.0 – 94.141.120.255)	Entire /24 block allocated to one organization — attacker controls the full range
Network Name	DGTL-NETWORK-2024-12-11	Name contains date 2024-12-11 — subnet created 7 days AFTER the attack on 2024-12-04
Subnet Created	2024-12-11T14:22:23Z	Critical finding: IP block registered AFTER the attack date — freshly provisioned or re-registered infrastructure; consistent with throwaway/bulletproof hosting
Route / ASN Created	2024-12-11T14:22:23Z via AS61087	BGP route advertisement also created 2024-12-11 — entire routing entry is post-attack; suggests infrastructure cleanup or re-registration after use
Organisation	DGTL TECH UK LLP (ORG-DTUL2-RIPE)	Small UK LIR (Local Internet Registry) — allocates its own IP space via RIPE, giving tenants reduced scrutiny compared to major cloud providers
Org Created	2021-01-20T10:18:39Z	Organisation pre-dates the attack by 3+ years — established entity providing hosting to potentially anonymous customers
Registered Address	71-75 Shelton Street, WC2H 9JQ, London, UK	Well-known virtual office address in Covent Garden — used by thousands of UK-registered shell companies and mailbox businesses; not a physical data centre
Country	GB (United Kingdom)	Geographic mismatch #3 — attacker claimed Turkish identity, sent from

WHOIS Field	Value	SOC Analysis / Significance
		US Pacific time zone infrastructure, but actual IP is UK-hosted
Abuse Contact	abuse@dgtl.tech	Provider-level abuse contact — small LIR with no major brand reputation risk; abuse reports may receive slow or no response
Status	ASSIGNED PA (Provider Aggregable)	IP space sub-allocated from DGTL TECH's parent block — attacker is a customer of this LIR, not a direct RIPE member
Last Modified	2026-04-06T13:13:36Z	WHOIS record recently updated — infrastructure still active; block not abandoned after the attack

2.3.1 The Story This WHOIS Data Tells

#	Intelligence Finding	What It Means
1	Subnet created 7 days after attack	The IP block 94.141.120.0/24 did not exist in RIPE until 2024-12-11 — 7 days after the December 4 attack. The attacker used infrastructure that was subsequently registered, suggesting either: (a) the subnet was freshly allocated post-attack as part of a cleanup/re-assignment, or (b) the attacker used pre-provisioned infrastructure that was only formally RIPE-registered after deployment. Either scenario points to deliberate infrastructure management.
2	Virtual office address	71-75 Shelton Street WC2H 9JQ is one of the UK's most commonly used registered-agent / virtual office addresses. DGTL TECH UK LLP operates as a small LIR from this address — a setup that offers legitimate IP allocation rights with minimal physical oversight.
3	Three-way geographic deception	The attacker constructed a false Turkish identity (supplier name, Turkish language, .com.tr domain) while operating from UK-hosted infrastructure (DGTL TECH, London) and using an automated sending system configured to US Pacific time (-0800). No element of the infrastructure matches the claimed origin.
4	Bulletproof-style hosting pattern	Small LIRs like DGTL TECH that allocate /24 blocks to anonymous customers are frequently used as bulletproof hosting — they handle abuse reports slowly or not at all, and customers can operate without standard KYC scrutiny required by major cloud providers (AWS, Azure, GCP).
5	AS61087 route advertisement	The BGP route for this /24 via AS61087 was also created on 2024-12-11. This means the network was not even globally routable before that date — the IP was reachable only because the sending was done via the provider's own routing before the RIPE route object was registered.

2.3.2 Geographic Mismatch Summary

Evidence Layer	Claimed / Apparent	Actual (Verified)	Mismatch
Sender identity	Turkish company (Acron Su ve Çevre Teknolojileri A.Ş.)	Unverified — no confirmed Turkish infrastructure	YES
Email timestamp Timezone	Turkish business hours expected (UTC+3)	US Pacific time (-0800 / UTC-8)	YES
Sending IP infrastructure	Turkey or EU expected	UK-hosted via DGTL TECH LLP, London (GB)	YES

WHOIS Output — Terminal Query: whois 94.141.120.32 (Full RIPE NCC Response)

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ whois 94.141.120.32
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html
%
% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.
%
% Information related to '94.141.120.0 - 94.141.120.255'
%
% Abuse contact for '94.141.120.0 - 94.141.120.255' is 'abuse@dgtl.tech'
inetnum:          94.141.120.0 - 94.141.120.255
netname:          DGTL-NETWORK-2024-12-11
org:              ORG-DTUL2-RIPE
country:          GB
admin-c:          DGTL6-RIPE
tech-c:           DGTL6-RIPE
status:           ASSIGNED PA
mnt-by:           MNT-DGTL
created:           2024-07-07T17:19:54Z
last-modified:    2026-04-06T13:13:36Z
source:           RIPE

organisation:     ORG-DTUL2-RIPE
org-name:         DGTL TECH UK LLP
org-type:         LIR
address:          71-75 Shelton Street
address:          WC2H 9JQ
address:          London
address:          UNITED KINGDOM
country:          GB
reg-nr:           0C434843
phone:            +442038079477
remarks:          Website: https://dgtl.tech/
admin-c:          DGTL6-RIPE
tech-c:           DGTL6-RIPE
abuse-c:          AR63612-RIPE
mnt-ref:          TNM-MNT
mnt-ref:          MNT-DGTL
mnt-by:           RIPE-NCC-HM-MNT
mnt-by:           MNT-DGTL
created:          2021-01-20T10:18:39Z
last-modified:    2026-05-13T07:27:49Z
```

```
org-name:      DGTL TECH UK LLP
org-type:      LIR
address:       71-75 Shelton Street
address:       WC2H 9JQ
address:       London
address:       UNITED KINGDOM
country:       GB
reg-nr:        0C434843
phone:         +442038079477
remarks:       Website: https://dgtl.tech/
admin-c:       DGTL6-RIPE
tech-c:        DGTL6-RIPE
abuse-c:       AR63612-RIPE
mnt-ref:       TNM-MNT
mnt-ref:       MNT-DGTL
mnt-by:        RIPE-NCC-HM-MNT
mnt-by:        MNT-DGTL
created:       2021-01-20T10:18:39Z
last-modified: 2026-05-13T07:27:49Z
source:        RIPE # Filtered

role:          DGTL LIR GROUP
address:       71-75 Shelton Street
address:       WC2H 9JQ
address:       London
address:       UNITED KINGDOM
nic-hdl:       DGTL6-RIPE
phone:         +44 20 3807 9477
mnt-by:        MNT-DGTL
created:       2020-02-06T20:02:30Z
last-modified: 2022-04-05T12:46:16Z
source:        RIPE # Filtered

% Information related to '94.141.120.0/24AS61087'

route:         94.141.120.0/24
origin:        AS61087
mnt-by:        MNT-DGTL
created:       2024-12-11T14:22:23Z
last-modified: 2024-12-11T14:22:23Z
source:        RIPE

% This query was served by the RIPE Database Query Service version 1.122.1 (DEXTER)
```

2.4 Email Content Analysis

Element	Finding
Subject	PURCHASE QUOTATION — generic but contextually appropriate for manufacturing industry
Body Language	Formal business tone with non-native grammar errors ('Dea Sir', 'We plan the purchase')
Embedded URLs	None — entirely attachment-based; bypasses URL reputation scanning and safe-link rewriting
Attachment Reference	'Please see technical specifications in the attached file' — direct instruction to open attachment

MIME Part Listing — emldump.py Output (All MIME Parts of the .eml File)

```
paul@kali: ~/SOC/Capstone Project/AgentTesla

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ python3 ../Tools/emldump.py 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
1: M      multipart/mixed
2:      923 text/html
3:      798831 application/x-tar (TECHNICAL SPECIFICATIONS.TAR)
```

Extracted Email Body Content — Decoded Plain-text Body from emldump.py

[illegible]

2.5 Social Engineering Assessment

Technique	Observation
Pretexting	Impersonates Acron Su ve Çevre Teknolojileri A.Ş.; real Turkish company with verifiable name, address, phone, and website; credibility is high
Authority	Sender presents as Purchase Manager, mid-level authority role that legitimises a quotation request without raising suspicion
Business Context Lure	Purchase quotation request is entirely routine in a manufacturing company, no unusual ask to trigger suspicion
No Urgency Tactic	Unlike many phishing emails this message uses no time pressure, it relies entirely on normalcy and routine business context
Grammar Anomalies	'Dea Sir' and 'We plan the purchase the equipment' non-native English; may increase perceived authenticity rather than alarm recipients
No Malicious URLs	No embedded hyperlinks entirely attachment-based; bypasses all URL reputation filters and safe-link rewriting services
Attachment as Sole Payload	All malicious content is in the attachment — the body exists only to make opening it seem like a routine business action

Technique	Observation
SECTION 3 ATTACHMENT ANALYSIS	

The attachment passed through three layers of deception before the executable payload was revealed. No execution was performed — all findings are based on static analysis only.

3.1 Deception / Attachment Chain

Layer	Appears To Be	Actually Is	MITRE Technique
1	TECHNICAL SPECIFICATIONS.TAR (application/x-tar)	RAR archive v4 (Win32)	T1036.008 — Masquerading: Wrong Extension
2	A document or specification file inside the archive	TECHNICAL SPECIFICATIONS.exe (Windows executable)	Extension confusion — spaces before .exe; Windows hides .exe by default
3	Unknown binary	PE32 .NET assembly — AgentTesla delivery format	T1027 — Obfuscated Files or Information (packed .NET payload)

Raw Attachment Extraction — emldump.py (Attachment Saved from .eml File)

```
(paul@kali) ~/SOC/Capstone Project/AgentTesla
$ python3 ../Tools/eloc.py 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
Extracted IP Addresses:
=====
94[.]141[.]120[.]32 - Roubaix, Hauts-de-France, FR, ISP: AS61087 DGTL TECH UK LLP

Extracted URLs:
=====
http://www[.]acronas[.]com[.]tr

Extracted Headers:
=====
Return-Path: <sertan@acronas.com.tr>
Authentication-Results: [recipient's email server]; dkim=none; dmarc=none; spf=softfail ([recipient's email server]: 94.141.120.32 is neither p
ermitted nor denied by domain of sertan@acronas.com.tr) smtp.mailfrom=sertan@acronas.com.tr
From: =?UTF-8?B?U2VydGFuIHOHT0tFUg==?= <sertan@acronas.com.tr>
To: [recipient's email address]
Subject: PURCHASE QUOTATION
Date: 4 Dec 2024 04:51:17 -0800
Message-ID: <20241204045117.4A43A7B93A5F2488@acronas.com.tr>

Extracted Attachments:
=====
Filename: TECHNICAL SPECIFICATIONS.TAR
MD5: b7635c9cc63619099419c68a2bf0d390
SHA1: 98683c9ee69dd591473629d231aacb1020db91b4
SHA256: 5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0
```

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ python3 ../Tools/emldump.py 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml -s 3 -d > TECHNICAL_SPECIFICATIONS.tar

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ ls -lh
total 1.9M
-rw-r--r-- 1 paul paul 1.1M May 18 18:54 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
-rw-r--r-- 1 paul paul 56K Dec 5 2024 2024-12-04-AgentTesla-variant-using-FTP.pcap
-rw-rw-r-- 1 paul paul 781K May 27 13:30 TECHNICAL_SPECIFICATIONS.tar

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$
```

True File Type Detection — file Command Output (TAR Extension Masking RAR Archive)

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ file TECHNICAL_SPECIFICATIONS.tar
TECHNICAL_SPECIFICATIONS.tar: RAR archive data, v4, os: Win32
```

RAR Archive Contents — unrar l Output (Payload Filename & Properties)

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ unrar l TECHNICAL_SPECIFICATIONS.tar

UNRAR 7.21 freeware      Copyright (c) 1993-2026 Alexander Roshal

Archive: TECHNICAL_SPECIFICATIONS.tar
Details: RAR 1.5

  Attributes      Size      Date      Time      Name
  -----
  ..A....      1096704   2024-12-04 00:02  TECHNICAL SPECIFICATIONS.exe
  -----
                   1096704                   1
```

3.2 File Properties

Property	Value
Attachment name (as declared in email)	TECHNICAL SPECIFICATIONS.TAR
Declared MIME type	application/x-tar
True container format	RAR archive data, v4, os: Win32
Payload filename inside archive	TECHNICAL SPECIFICATIONS.exe
Payload size	1,096,704 bytes (1.05 MB)
Payload file type	PE32 executable (GUI) Intel 80386, Mono/.NET assembly, for MS Windows — 3 sections
Compilation timestamp	2024-12-04 00:02 — compiled same day as delivery
.NET assembly confirmed	Yes — MZ magic bytes + _CorExeMain signature detected

3.3 File Hash Values

File	Algorithm	Hash Value
RAR Container (TECHNICAL_SPECIFICATIO NS.tar)	MD5	b7635c9cc63619099419c68a2bf0d390
RAR Container (TECHNICAL_SPECIFICATIO NS.tar)	SHA256	5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915a c336b6f1d59f0
TECHNICAL SPECIFICATIONS.exe	MD5	65feefe926eb3f734b6968b35c23acb3
TECHNICAL SPECIFICATIONS.exe	SHA256	d1b068b826e3a9527cddd09866886caba895f390af930a9b35 c027eb1c2db34c

Hash Generation — md5sum / sha256sum Output for Both Files

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ unrar e TECHNICAL_SPECIFICATIONS.tar "TECHNICAL SPECIFICATIONS.exe" .

UNRAR 7.21 freeware      Copyright (c) 1993-2026 Alexander Roshal

Extracting from TECHNICAL_SPECIFICATIONS.tar

Extracting ./TECHNICAL_SPECIFICATIONS.exe          OK
All OK

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ file "TECHNICAL_SPECIFICATIONS.exe"
TECHNICAL_SPECIFICATIONS.exe: PE32 executable for MS Windows 4.00 (GUI), Intel i386 Mono/.Net assembly, 3 sections

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ echo "MD5 hashes:"
md5sum TECHNICAL_SPECIFICATIONS.tar "TECHNICAL SPECIFICATIONS.exe"

echo ""
echo "SHA256 hashes:"
sha256sum TECHNICAL_SPECIFICATIONS.tar "TECHNICAL SPECIFICATIONS.exe"

MD5 hashes:
b7635c9cc63619099419c68a2bf0d390  TECHNICAL_SPECIFICATIONS.tar
65feefe926eb3f734b6968b35c23acb3  TECHNICAL_SPECIFICATIONS.exe

SHA256 hashes:
5c98308c69c84a57214442e2cad9f8f0fcd8bab8e6050f9915ac336b6f1d59f0  TECHNICAL_SPECIFICATIONS.tar
d1b068b826e3a9527cddd09866886caba895f390af930a9b35c027eb1c2db34c  TECHNICAL_SPECIFICATIONS.exe
```

3.4 Malware Indicators (Static String Analysis: No Execution)

Static string analysis of the PE binary revealed the following credential-harvesting indicators embedded in the .NET assembly:

Indicator String	Significance
pbHidePassword	UI element for masking a password input field; credential capture form component
txtPassword	Password text box control, input field for capturing credentials
pbShowPassword_Click	Show/hide password toggle event handler, interactive credential harvesting UI
set_PasswordChar	.NET method for concealing typed characters, consistent with fake login form overlay

These strings are consistent with a credential harvesting component — either a fake login form overlay or a browser password vault extractor. Combined with the .NET assembly delivery method, this is consistent with the AgentTesla infostealer family.

Static String Analysis — Malware Indicator Strings in PE Binary

```
(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ strings "TECHNICAL SPECIFICATIONS.exe" > strings.txt

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ ls -lh
total 3.0M
-rw-r--r-- 1 paul paul 1.1M May 18 18:54 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
-rw-r--r-- 1 paul paul 56K Dec 5 2024 2024-12-04-AgentTesla-variant-using-FTP.pcap
drwxrwxr-x 2 paul paul 4.0K May 29 11:23 ftp_exports
-rw-rw-r-- 1 paul paul 55K Jun 1 21:31 strings.txt
-rw-rw-r-- 1 paul paul 1.1M Dec 4 2024 'TECHNICAL SPECIFICATIONS.exe'
-rw-rw-r-- 1 paul paul 781K Jun 1 21:26 TECHNICAL_SPECIFICATIONS.tar

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ cat strings.txt
!This program cannot be run in DOS mode.
.text
.rsrc
@.reloc
...

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ grep -Ei "user|pass|login|pwd|credential" strings.txt

pbHidePassword
txtPassword
pbShowPassword
txtUsername
DebuggerNonUserCodeAttribute
pbHidePassword_Click
pbShowPassword_Click
btnLogin_Click
frmLogin
btnLogin
set_PasswordChar
PiggyBank.frmLogin.resources

(paul@kali)-[~/SOC/Capstone Project/AgentTesla]
$ grep -Ei "ConvertHexString|Base64|Decode|Decrypt" strings.txt

<ConvertHexStringToByteArray>b__0
ConvertHexStringToByteArray
```

3.5 VirusTotal Hash Reputation Analysis

File	VirusTotal Result	Notable Vendor Detection
TECHNICAL_SPECIFICATIONS.tar (RAR container)	MALICIOUS: Trojan/AgentTesla	Antiy-AVL: Trojan/Win32.AgentTesla (Keylogger)
TECHNICAL SPECIFICATIONS.exe	MALICIOUS: Trojan.msil/Strictor	Multiple vendors: confirmed infostealer payload

VirusTotal Analysis — SHA256 of RAR Container (Trojan/AgentTesla Detection)

5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0

49

63

Community Score

13

49/63 security vendors flagged this file as malicious

5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0

TECHNICALXSPECIFICATIONS.TAR

Size780.11 KB

Last Analysis Date1 month ago

RAR

rar

persistence

detect debug environment

contains-pe

checks-bios

checks-network adapters

attachment

calls-wmi

spreader

clipboard

checks-user-input

long sleeps

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY10

Join our Community

and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label

trojan.msil/agenttesla

Threat categories

trojan

Family labels

msil

agenttesla

crypt

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan/Win.MalwareX-gen.C5702736	AliCloud	Trojan(spy):MSIL/Noon.gyf
ALYac	Gen:Variant.Application.Strictor.240530	Antiy-AVL	Trojan/MSIL.SnakeKeylogger
Avast	Trojan.Generic.MD79AA95	Avast	Win32-MalwareX-gen [Misc]



SECTION 4 | INDICATORS OF COMPROMISE (IOC) TABLE

4.1 Email Header IOCs

IOC Type	Value	Source	Confidence
Sender email address	sertan@acronas.com.tr	Email header From:	High
Sending domain	acronas.com.tr	Email header	High
Originating IP address	94.141.120.32	Email header Received:	High
SPF result	softfail	MX Toolbox analysis	High
DKIM result	none — no signature	MX Toolbox analysis	High
DMARC result	none — no policy	MX Toolbox analysis	High
Timezone anomaly	-0800 (US Pacific) from Turkish sender	Header Date: field	High

4.2 Infrastructure Intelligence IOCs (WHOIS — 94.141.120.32)

IOC Type	Value	Source	Confidence
IP subnet	94.141.120.0/24	RIPE WHOIS query	High
Subnet name	DGTL-NETWORK-2024-12-11	RIPE WHOIS — netname	High
Subnet creation date	2024-12-11 — 7 days AFTER the attack	RIPE WHOIS — created field	High
ASN / Route	AS61087 — route created 2024-12-11	RIPE WHOIS — route object	High
Hosting Organisation	DGTL TECH UK LLP (ORG-DTUL2-RIPE)	RIPE WHOIS — org field	High
Registered address	71-75 Shelton Street, WC2H 9JQ, London, UK	RIPE WHOIS — address	High
Abuse contact	abuse@dgtl.tech	RIPE WHOIS — abuse-c	High
Geographic mismatch	Claimed Turkey / Timezone US Pacific / IP hosted UK	Cross-referencing header + WHOIS	High

IOC Type	Value	Source
Declared attachment name	TECHNICAL SPECIFICATIONS.TAR	emldump.py
True container format	RAR archive data, v4, os: Win32	file command (magic bytes)
Evasion technique	File masquerading — T1036.008	file command vs declared MIME type
Payload filename	TECHNICAL SPECIFICATIONS.exe	unrar l
Payload file type	PE32 .NET executable (GUI) for MS Windows	file command
Compilation timestamp	2024-12-04 00:02 (same day as delivery)	PE header compilation date
Embedded URLs	None — attachment-only attack	Email body analysis
VirusTotal classification	Trojan/AgentTesla (RAR)Trojan.msil/Strictor (EXE)	VirusTotal hash lookup

4.4 File Hash IOCs

File	Algorithm	Hash Value
RAR container	MD5	b7635c9cc63619099419c68a2bf0d390
RAR container	SHA256	5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0
TECHNICAL SPECIFICATIONS.exe	MD5	65feefe926eb3f734b6968b35c23acb3
TECHNICAL SPECIFICATIONS.exe	SHA256	d1b068b826e3a9527cddd09866886caba895f390af930a9b35c027eb1c2db34c

SECTION 5 | RISK RATING

OVERALL RISK RATING

HIGH

Confirmed executable payload delivered. All authentication controls failed. File masquerading employed. Full credential exfiltration confirmed in Phase 2.

5.1 Risk Dimension Breakdown

Overall Severity	HIGH	Confirmed executable payload successfully delivered; all authentication controls absent or failed; file masquerading employed to evade filters
Delivery Method	HIGH	File masquerading (.TAR → RAR → .EXE) bypasses extension-based email security; no malicious URLs to flag by reputation scanning
Payload Capability	HIGH	PE32 .NET binary with credential-harvesting strings; consistent with AgentTesla — keylogging, browser credential theft, and FTP exfiltration
Authentication Failures	HIGH	SPF softfail, DKIM none, DMARC none — domain has no controls that allow detection or rejection of spoofed email
Social Engineering	MEDIUM	Plausible business purchase pretext using real supplier branding; grammar anomalies reduce sophistication but increase perceived authenticity for non-native speakers
Victim Exposure	HIGH	Manufacturing company routinely receives supplier quotation requests — victim profile is well-matched to the lure; low suspicion threshold

SECTION 6 | INITIAL HYPOTHESIS OF ATTACK

The threat actor conducted a targeted malspam campaign against Globex Manufacturing Ltd., selecting the purchase quotation pretext as a high-credibility lure appropriate for a manufacturing company. The campaign delivered the AgentTesla infostealer via a multi-layer file masquerading chain designed to evade common email security controls.

6.1 Reconstructed Attack Chain

#	Stage	Action / Method	Evidence
1	Target Selection	Manufacturing company identified routinely receives supplier quotation emails; low suspicion profile	Email subject PURCHASE QUOTATION; lure content matches industry
2	Infrastructure Setup	Attacker provisioned or used UK-hosted IP 94.141.120.32 via DGTL TECH UK LLP; a small London LIR operating from a virtual office address; subnet DGTL-NETWORK-2024-12-11 registered 7 days post-attack suggesting throwaway infrastructure	RIPE WHOIS — subnet created 2024-12-11; virtual office at 71-75 Shelton Street WC2H 9JQ
3	Sender Preparation	Email sent from sertan@acronas.com.tr via IP 94.141.120.32 using US Pacific time zone (-0800) while claiming Turkish origin: three-way geographic mismatch: Turkey claimed / US Pacific time zone / UK-hosted IP	SPF softfail; direct single-hop IP; WHOIS country Great Britain vs claimed Turkey
4	Payload Preparation	AgentTesla compiled same day as delivery (2024-12-04 00:02); packed into RAR and renamed with .TAR extension	Compilation timestamp; file masquerading chain confirmed by file command
5	Email Delivery	Single-hop delivery to victim, no relay chain anonymization used	Received: header- single hop from 94.141.120.32

6.2 MITRE ATT&CK Mapping

Tactic	Technique	ID	Evidence
Initial Access	Spear Phishing Attachment	T1566.001	Email with malicious RAR attachment disguised as supplier technical document
Execution	User Execution: Malicious File	T1204.002	Victim required to open the attachment and execute the .exe payload
Defense Evasion	Masquerading: Wrong Extension	T1036.008	.TAR extension on RAR archive; .exe inside, bypasses extension-based email filters
Defense Evasion	Obfuscated Files or Information	T1027	.NET binary likely packed to evade static signature detection

6.3 Confidence Assessment

Finding	Confidence	Basis
File masquerading chain and authentication failures	HIGH	Direct evidence from raw .eml file — confirmed by file command and MX Toolbox
WHOIS infrastructure intelligence — UK-hosted bulletproof-style provider	HIGH	RIPE NCC WHOIS confirms subnet DGTL-NETWORK-2024-12-11 created 7 days post-attack; virtual office address; three-way geographic mismatch confirmed
Payload type (.NET assembly with credential-harvesting strings)	HIGH	Confirmed by static string analysis, no execution required
AgentTesla family attribution	HIGH	VirusTotal hash lookup confirms Trojan/AgentTesla and Trojan.msil/Strictor classifications